

Discrete Structures

Lecture # 08

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EXERCISE

Define a binary relation **E** on the set of the integers **Z**, as follows:

$m, n \in \mathbb{Z}, m \mathbf{E} n \Leftrightarrow m - n$ is even.

a. Is **0E0**?

Is **5E2**?

Does $(6,6) \in \mathbf{E}$?

Does $(-1,7) \in \mathbf{E}$?

b. Prove that for any even integer n , **nE0**.

SOLUTION

b. For any **even integer**, n , we have

$$n - 0 = n, \quad \text{an **even integer**}$$

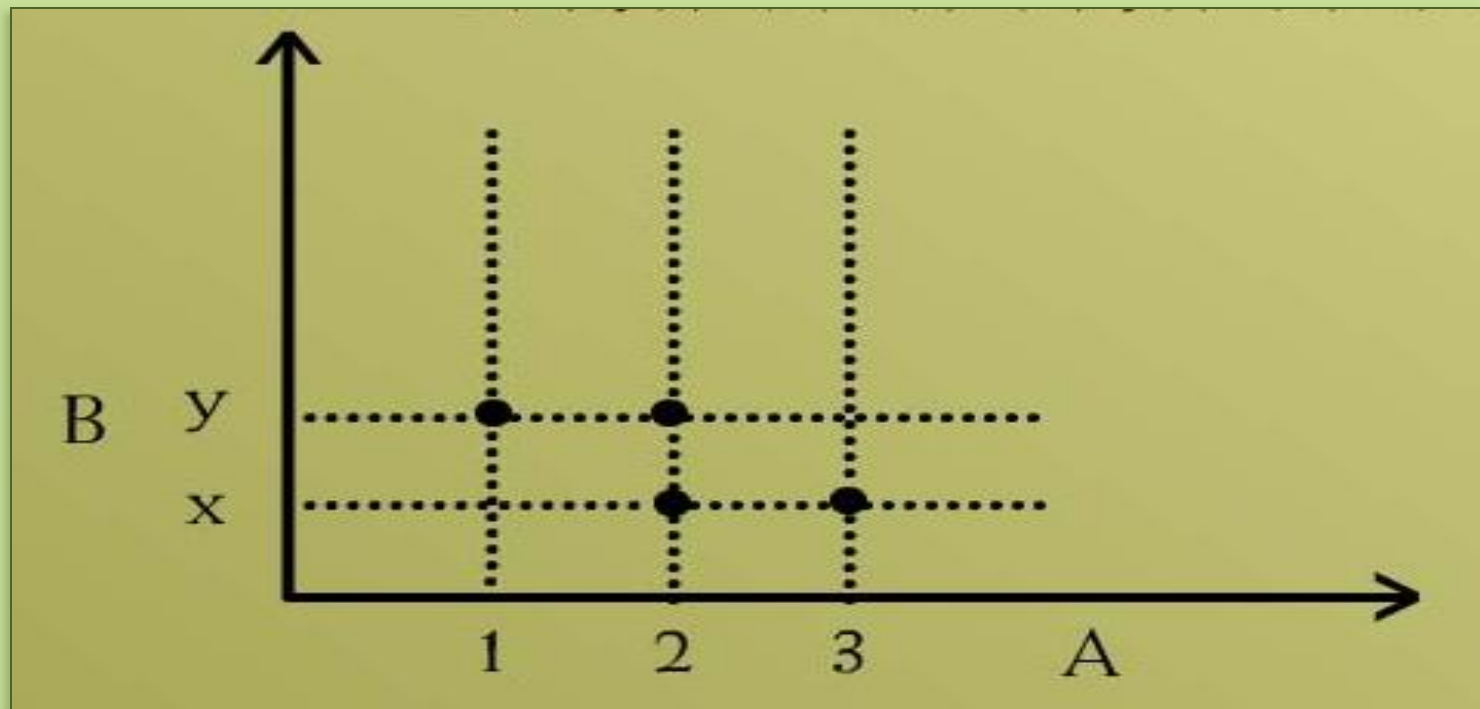
$$\text{so } (n, 0) \in E$$

$$\text{or equivalently } n \in 0$$

CARTESIAN DIAGRAM OF A RELATION

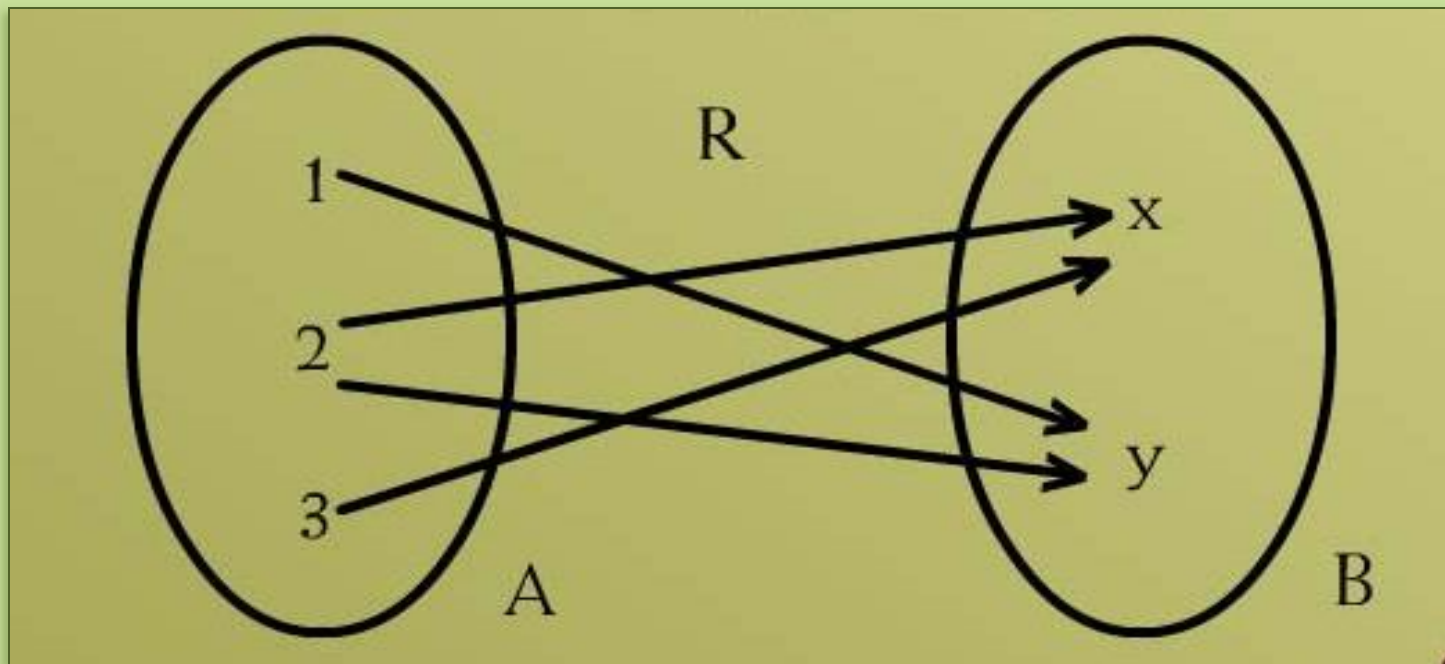
Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$

$R = \{ (1, y) , (2, x) , (2, y) , (3, x) \}$



ARROW DIAGRAM OF A RELATION

Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$
 $R = \{(1, y), (2, x), (2, y), (3, x)\}$

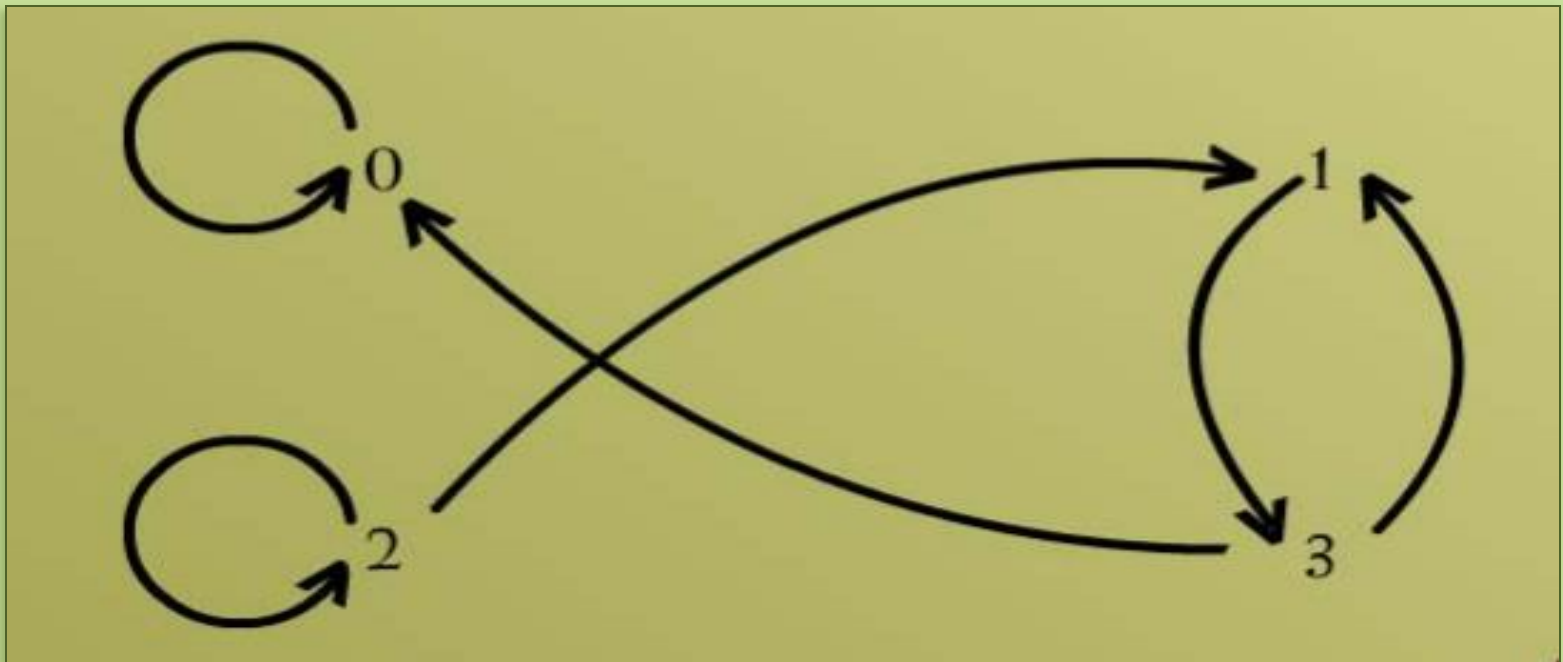


DIRECTED GRAPH OF A RELATION

Let $A = \{0, 1, 2, 3\}$

$A \times A$

$R = \{ (0,0) , (1, 3) , (2, 1) , (2, 2) , (3, 0) , (3, 1) \}$



MATRIX REPRESENTATION OF A RELATION

Let $A = \{a_1, a_2, \dots, a_n\}$

$B = \{b_1, b_2, \dots, b_m\}$

Let R be a relation from A to B .

Define the matrix M of order $n \times m$ by

$$M_{(i,j)} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$$

for $i=1,2,\dots,n$ and $j=1,2,\dots,m$

EXAMPLE

Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$

$R = \{(1,y), (2,x), (2,y), (3,x)\}$

Order of matrix = 3×2

$$M = \begin{matrix} & \begin{matrix} x & y \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \end{matrix} \quad 3 \times 2$$

**Define
A x B
first**

EXAMPLE

For the relation matrix

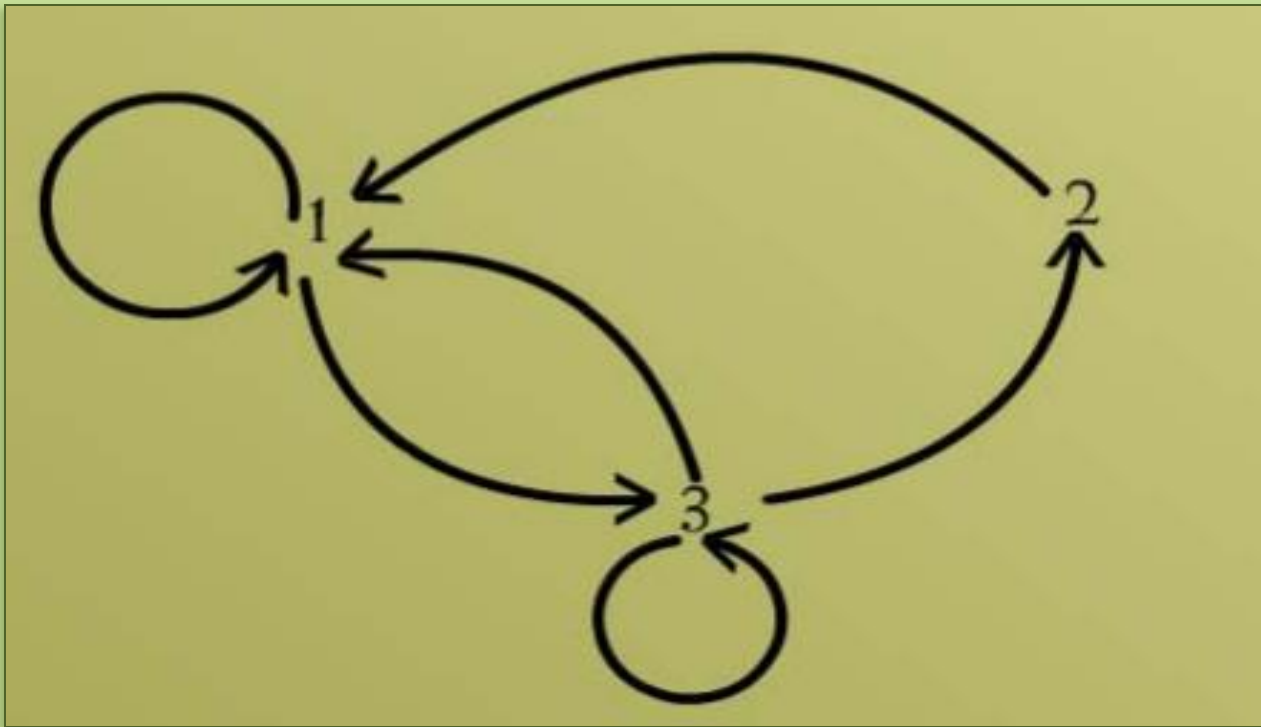
$$\mathbf{M} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

1. List set of ordered pairs represented by $\mathbf{M}$.
2. Draw the directed graph of the relation.

Solution contd...

$$M = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

1. $R = \{ (1, 1), (1, 3), (2, 1), (3, 1), (3, 2), (3, 3) \}$



EXERCISE

Let $A = \{2, 4\}$ and $B = \{6, 8, 10\}$
and define relations R and S from A to B as follows:

$$R = \{(x, y) \in A \times B \mid x R y \Leftrightarrow x \mid y\}$$

$$S = \{(x, y) \in A \times B \mid x S y \Leftrightarrow y - 4 = x\}$$

State explicitly which ordered pairs are in
 $A \times B$, R , S , $R \cup S$ and $R \cap S$

SOLUTION

$$A \times B$$

$$= \{(2,6), (2,8), (2,10), (4,6), (4,8), (4,10)\}$$

$$R = \{(2,6), (2,8), (2,10), (4,8)\}$$

$$S = \{(2,6), (4,8)\}$$

$$S \subseteq R$$

$$R \cup S = \{(2,6), (2,8), (2,10), (4,8)\} = R$$

$$R \cap S = \{(2,6), (4,8)\} = S$$

REFLEXIVE RELATION

Let R be a relation on a set A . R is reflexive if, and only if, for all $a \in A$, $(a, a) \in R$. Or equivalently aRa . That is, each element of A is related to itself.

REFLEXIVE RELATION

REMARK:

R is not reflexive iff there is an element " a " in A such that $(a, a) \notin R$. That is, some element " a " of A is not related to itself.

EXAMPLE

Let $A = \{1, 2, 3, 4\}$

We define relations R_1, R_2, R_3, R_4 on A as follows:

$$R_1 = \{(1, 1), (3, 3), (2, 2), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 4), (2, 2), (3, 3), (4, 3)\}$$

$$R_3 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_4 = \{(1, 3), (2, 2), (2, 4), (3, 1), (4, 4)\}$$

EXAMPLE

$$R_1 = \{(1, 1), (3, 3), (2, 2), (4, 4)\}$$

R_1 is reflexive, since $(a, a) \in R_1$ for all $a \in A$.

$$R_2 = \{(1, 1), (1, 4), (2, 2), (3, 3), (4, 3)\}$$

R_2 is not reflexive, because $(4, 4) \notin R_2$.

EXAMPLE

$$R_3 =$$

$$\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

R_3 is reflexive, since $(a, a) \in R_3$ for all $a \in A$.

$$R_4 = \{(1, 3), (2, 2), (2, 4), (3, 1), (4, 4)\}$$

R_4 is not reflexive, because $(1, 1) \notin R_4$,
 $(3, 3) \notin R_4$

DIRECTED GRAPH OF A REFLEXIVE RELATION

Let $A = \{1, 2, 3, 4\}$

$R_1 = \{(1, 1), (3, 3), (2, 2), (4, 4)\}$



R_1 is reflexive

DIRECTED GRAPH OF A REFLEXIVE RELATION

Let $A = \{1, 2, 3, 4\}$

$$R_2 = \{(1, 1), (1, 4), (2, 2), (3, 3), (4, 3)\}$$



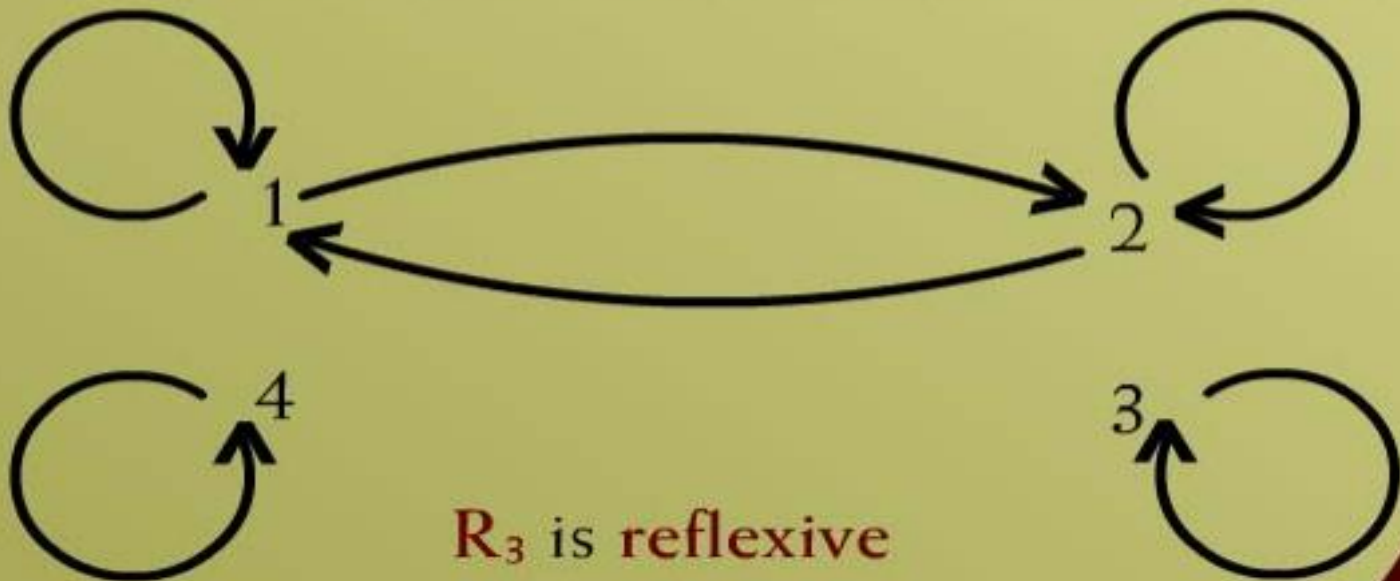
R_2 is not **reflexive**, as there is no loop at 4.

DIRECTED GRAPH OF A REFLEXIVE RELATION

Let

$$A = \{1, 2, 3, 4\}$$

$$R_3 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$



DIRECTED GRAPH OF A REFLEXIVE RELATION

Let

$$A = \{1, 2, 3, 4\}$$

$$R_4 = \{(1, 3), (2, 2), (2, 4), (3, 1), (4, 4)\}$$



R_4 is not **reflexive**, as there are **no loops** at **1 and 3**.

MATRIX REPRESENTATION OF A REFLEXIVE RELATION

Let $A = \{a_1, a_2, \dots, a_n\}$. A Relation R on A is reflexive if and only if $(a_i, a_i) \in R \ \forall \ i = 1, 2, \dots, n$.

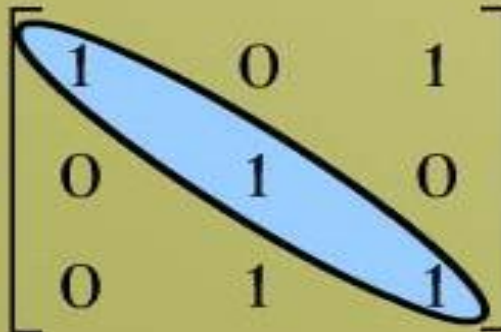
Accordingly, R is reflexive if all the elements on the main diagonal of the matrix M representing R are equal to 1.

EXAMPLE

Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,2), (3,3)\}$

R is reflexive

$$M = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{matrix}$$


SYMMETRIC RELATION

Let R be a relation on a set A . R is symmetric **if, and only if**, for all $a, b \in A$, if $(a, b) \in R$ then $(b, a) \in R$. That is, if aRb then bRa .

REMARK:

R is not symmetric **iff** there are elements a and b in A such that $(a, b) \in R$ but $(b, a) \notin R$.

EXAMPLE

Let $A = \{1, 2, 3, 4\}$

R_1, R_2, R_3, R_4

$$R_1 = \{(1, 1), (1, 3), (2, 4), (3, 1), (4, 2)\}$$

R_1 is symmetric.

$$R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

R_2 is symmetric.

EXAMPLE

$$R_3 = \{(2, 2), (2, 3), (3, 4)\}$$

R_3 is not **symmetric**, because $(2,3) \in R_3$
but $(3,2) \notin R_3$.

$$R_4 = \{(1, 1), (2, 2), (3, 3), (4, 3), (4, 4)\}$$

R_4 is not **symmetric**, because $(4,3) \in R_4$
but $(3,4) \notin R_4$.

DIRECTED GRAPH OF A SYMMETRIC RELATION

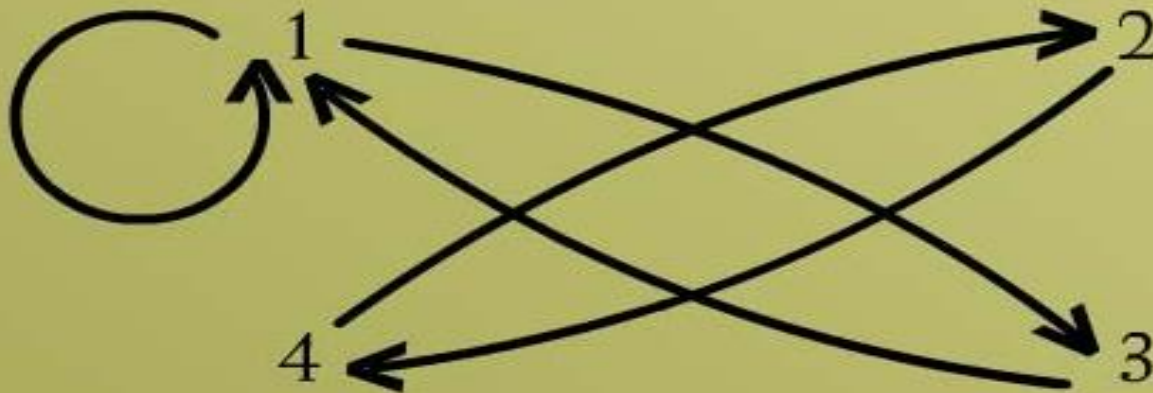
For a **symmetric** directed **graph** whenever there is **an arrow** going from **one point** of the **graph** to a **second**, there is **an arrow** going from the **second point** back to the **first**.

EXAMPLE

Let

$$A = \{1, 2, 3, 4\}$$

$$R_1 = \{(1, 1), (1, 3), (2, 4), (3, 1), (4, 2)\}$$



R_1 is symmetric

EXAMPLE

Let $A = \{1, 2, 3, 4\}$

$$R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$



R_2 is symmetric.

EXAMPLE

Let $A = \{1, 2, 3, 4\}$

$$R_3 = \{(2, 2), (2, 3), (3, 4)\}$$



R_3 is not symmetric since there are arrows from 2 to 3 and from 3 to 4 but not conversely.

EXAMPLE

Let $A = \{1, 2, 3, 4\}$

$$R_4 = \{(1, 1), (2, 2), (3, 3), (4, 3), (4, 4)\}$$



R_4 is not symmetric since there is an arrow from 4 to 3 but no arrow from 3 to 4

MATRIX REPRESENTATION OF A SYMMETRIC RELATION

Let $A = \{a_1, a_2, \dots, a_n\}$. A relation R on A is symmetric if and only if for all $a_i, a_j \in A$, if $(a_i, a_j) \in R$ then $(a_j, a_i) \in R$.

Accordingly, R is symmetric if the elements in the i th row are the same as the elements in the i th column of the matrix M representing R .

EXAMPLE

Let $A = \{1, 2, 3\}$

$R = \{(1, 3), (2, 2), (3, 1), (3, 3)\}$

$$M = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

M is symmetric i.e. $M = M^t$

Transitive Relation

Let R be a relation on a set A .
 R is transitive if and only if
for all $a, b, c \in A$, if $(a, b) \in R$
and $(b, c) \in R$ then $(a, c) \in R$.

That is, if aRb and bRc then aRc .

EXAMPLE

In words, if any one element is related to a second and that second element is related to a third, then the first is related to the third.

REMARKS:

R is not transitive iff there are elements a, b, c in A such that $(a,b) \in R$ and $(b,c) \in R$ but $(a,c) \notin R$.

EXAMPLE

Let $A = \{1, 2, 3, 4\}$ define relations.

$$R_1 = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$$

$$R_2 = \{(1, 2), (1, 4), (2, 3), (3, 4)\}$$

$$R_3 = \{(2, 1), (2, 4), (2, 3), (3, 4)\}$$

EXAMPLE

$$R_1 = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$$

R_1 is transitive. Because $(1,2) \in R_1$ and $(2,3) \in R_1 \Rightarrow (1,3) \in R_1$

$$R_2 = \{(1, 2), (1, 4), (2, 3), (3, 4)\}$$

R_2 is not transitive. Because $(1,2) \in R_2$ and $(2,3) \in R_2$ but $(1,3) \notin R_2$

$$R_3 = \{(2, 1), (2, 4), (2, 3), (3,4)\}$$

R_3 is transitive. Because $(2,3) \in R_3$ and $(3,4) \in R_3 \Rightarrow (2,4) \in R_3$

DIRECTED GRAPH OF A TRANSITIVE RELATION

For a **transitive** directed **graph**, whenever there is **an arrow** going from **one point** to the **second**, and from the **second** to the **third**, there is **an arrow** going directly from the **first** to the **third**.

EXAMPLE

Let $A = \{1, 2, 3, 4\}$

$R_1 = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$



R_1 is transitive.

EXAMPLE

Let $A = \{1, 2, 3, 4\}$

$$R_2 = \{(1, 2), (1, 4), (2, 3), (3, 4)\}$$



R_2 is not **transitive** since there is **an arrow** from **1** to **2** and from **2** to **3** but **no arrow** from **1** to **3** directly.

EXAMPLE

Let $A = \{1, 2, 3, 4\}$

$R_3 = \{(2, 1), (2, 4), (2, 3), (3, 4)\}$



R_3 is transitive

Matrix Representation of Transitive Relation

- Let A be the given matrix.
- Identify all zeros of the A .
- Take the square of the given matrix A .
- The relation is transitive if and only if the squared matrix has no nonzero entry where the original had a zero.

Example

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Let A be the given matrix.

Now

$$A^2 = A.A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

As all zeros in A^2 are intact i.e. zeros in A^2 are at the same position as that of A . Therefore the given relation is transitive.

EXERCISE

Let $A = \{0, 1, 2\}$

$R = \{(0,2), (1,1), (2,0)\}$

1. Is R reflexive? Symmetric? Transitive?
2. Which **ordered pairs** are needed in R to make it a **reflexive** and **transitive relation**.

SOLUTION

$$R = \{(0,2), (1,1), (2,0)\}$$

1. **R** is not **reflexive**, since $0 \in A$ but $(0, 0) \notin R$.

R is **symmetric**.

Because $(0,2) \in R \Rightarrow (2,0) \in R$

R is not **transitive**, since $(0, 2)$ & $(2, 0) \in R$ but $(0, 0) \notin R$.

SOLUTION

$$R = \{(0,2), (1,1), (2,0)\}$$

1. R is not reflexive.

R to be reflexive it must contain $(0, 0)$ and $(2, 2)$.

2. R is not transitive.

For R to be transitive it must contain $(0, 0)$ and $(2, 2)$.

EXERCISE

Define a relation **L** on the set of **real numbers R** be defined as follows:

for all $x, y \in \mathbf{R}$, $x \mathbf{L} y \Leftrightarrow x < y$.

- a. Is **L** reflexive?
- b. Is **L** symmetric?
- c. Is **L** transitive?

R here is representing Real Numbers and L is representing relation.

SOLUTION

$$x \mathbf{L} y \Leftrightarrow x < y$$

- a. **L** is not **reflexive**, because $x < x$ for any **real number** x . is not true.

For example $1 \nless 1$

- b. **L** is not **symmetric**, because for all $x, y \in \mathbb{R}$, if $x < y$ then $y < x$ which can't be true at the same time.

For example $1 < 2$ and $2 \nless 1$

SOLUTION

$$x \mathbf{L} y \Leftrightarrow x < y$$

- c. **L** is **transitive**, because for all, $x, y, z \in \mathbf{R}$, if $x < y$ and $y < z$, then $x < z$.
(by **transitive law** of order of **real numbers**).

Note: These properties are independent of each other.

EXAMPLE

Define a relation **R** on the set of **positive integers** $\mathbb{Z}^+$ as follows:

for all $a, b \in \mathbb{Z}^+$, **a R b** iff **a × b** is **odd**.

Determine whether the relation is.

- a. **Reflexive**
- b. **Symmetric**
- c. **Transitive**

SOLUTION

For reflexivity, we have to show that, aRa iff $a \times a = \text{odd number}$.

And it contradicts the definition of reflexivity. Because $2R2 \leftrightarrow 2 \times 2 = 4$ which is not odd.

Hence, R is not reflexive.

SOLUTION

b. Symmetric

R is symmetric.

if $a R b$ then $a \times b$ is odd
or equivalently $b \times a$ is odd

$$\therefore (b \times a = a \times b)$$

$$\Rightarrow b R a.$$

SOLUTION

c. **Transitive**

R is transitive.

if **a R b** then **a × b** is odd

$\Rightarrow$ both "**a**" and "**b**" are odd.

b R c then **b × c** is odd

$\Rightarrow$ both "**b**" and "**c**" are odd.

$\Rightarrow$ **a R c** because **a × c** is odd.

EXAMPLE

Let "**D**" be the "**divides**" relation on **Z** defined as:

$$\text{for all } m, n \in \mathbf{Z}, m \mathbf{D} n \Leftrightarrow m \mid n$$

Determine whether **D** is **reflexive**, **symmetric** or **transitive**. Justify your answer.

Note: Here the set of integers do not include 0 i.e. $\mathbf{Z} - \{0\}$

SOLUTION

Reflexive:

D is clearly reflexive.

Let $m \in \mathbb{Z}$, since every integer divides itself so $m \mid m \quad \forall m \in \mathbb{Z}$

therefore $m D m \quad \forall m \in \mathbb{Z}$

For example $2 \mid 2$

SOLUTION

Symmetric:

This **relation** is not **symmetric**.

For example 2 **divides** 6.

But 6 does not **divides** 2.

i.e **$2D6$** but **$6 \nmid 2$** .

SOLUTION

Transitive:

If m **divides** n and n **divides** p .
Then m **divides** p .

EQUIVALENCE RELATION

Let A be a non-empty set and R a binary relation on A . R is an equivalence relation if and only if, R is reflexive, symmetric, and transitive.

EXAMPLE

Let

$$A = \{1, 2, 3, 4\}$$

$$R = \{ (1, 1) , (2, 2) , (2, 4) , (3, 3) , (4, 2) , (4, 4) \}$$

R is reflexive, symmetric and transitive.

CONGRUENCES RELATION

Let m and n be integers and d be a positive integer.

The notation $m \equiv n \pmod{d}$
means that

$d \mid (m - n)$ { d divides m minus n }

$\Leftrightarrow$ There exists an integer k such that

$$(m - n) = d \cdot k$$

EXAMPLE

- a. Is $22 \equiv 1 \pmod{3}$? b. Is $-5 \equiv +10 \pmod{3}$?
c. Is $7 \equiv 7 \pmod{4}$? d. Is $14 \equiv 4 \pmod{4}$?

Solution:

a. Since $22 - 1 = 21$

21 is **divisible** by 3

Hence $3 \mid (22 - 1)$, and so $22 \equiv 1 \pmod{3}$

b. Since $-5 - 10 = -15$

-15 is **divisible** by 3

Hence $3 \mid ((-5) - 10)$, and so $-5 \equiv 10 \pmod{3}$

Solution contd...

c. Since $7 - 7 = 0$

Hence $4 \mid (7-7)$, and so $7 \equiv 7 \pmod{4}$

d. Since $14 - 4 = 10$ but 4 does not divide 10

Hence $14 \not\equiv 4 \pmod{4}$.

EXERCISE

Define a relation **R** on the set of all integers **Z** as follows:

for all integers **m** and **n**,

$$m \mathbf{R} n \Leftrightarrow m \equiv n(\text{mod } 3)$$

Prove that **R** is an **equivalence relation**.

SOLUTION

R is reflexive.

Every element is related to itself because

$$m - m = 0$$

Zero is divisible by every number.

Hence $3 \mid (m-m)$, and so **$m \equiv m \pmod{3}$**

SOLUTION

R is symmetric.

We have to show that
if $m R n$ then $n R m$.

$$\begin{aligned} m R n &\Rightarrow m \equiv n \pmod{3} \\ &\Rightarrow 3 \mid (m-n) \\ &\Rightarrow m-n = 3k, \text{ for some integer } k. \\ &\Rightarrow n-m = 3(-k), -k \in \mathbb{Z} \\ &\Rightarrow 3 \mid (n-m) \\ &\Rightarrow n \equiv m \pmod{3} \\ &\Rightarrow n R m \end{aligned}$$

SOLUTION

R is transitive.

mRn and **nRp** means

$$m \equiv n \pmod{3} \text{ and } n \equiv p \pmod{3}$$

$$\Rightarrow 3 \mid (m-n) \text{ and } 3 \mid (n-p)$$

$$\Rightarrow (m-n) = 3r \text{ and } (n-p) = 3s$$

for some $r, s \in \mathbb{Z}$

Adding these two equations, we get,

$$(m-n) + (n-p) = 3r + 3s$$

$$\Rightarrow m-p = 3(r+s), \text{ where } r+s \in \mathbb{Z}$$

$$\Rightarrow 3 \mid (m-p)$$

$$\Rightarrow m \equiv p \pmod{3} \Leftrightarrow mRp$$